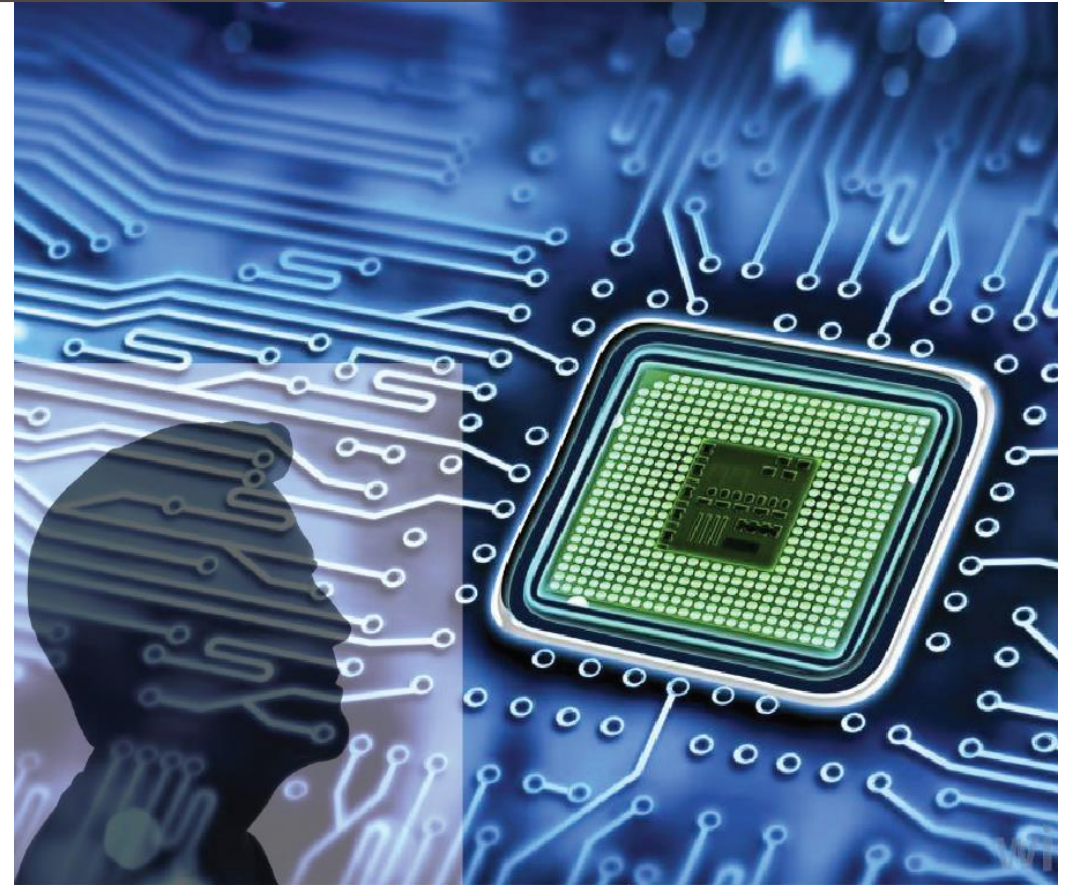


VLSI DESIGN FLOW

Week-11 Lecture-1 DFT

Sneh Saurabh
4th November, 2024



VDF: Revision/Announcements

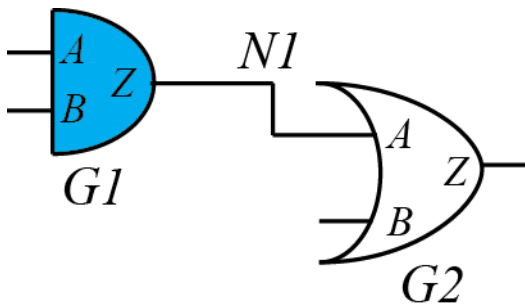
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Stuck-at Faults: Fault-site

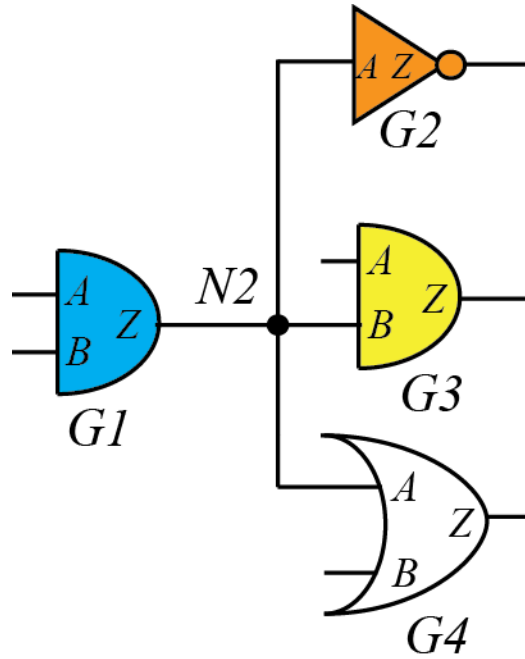
Fault-site:

- The point where a fault exists or we assume it to exist
- Emulate a stuck-at fault by first disconnecting the corresponding signal source.
 - Tie it to the constant logic (either 0 or 1 depending on the fault type)



Fanout-free net:

- 1 fault site

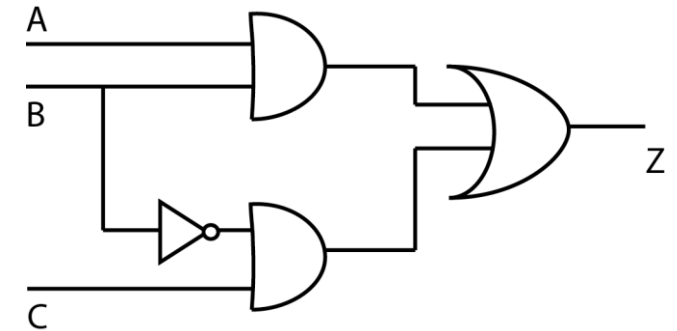
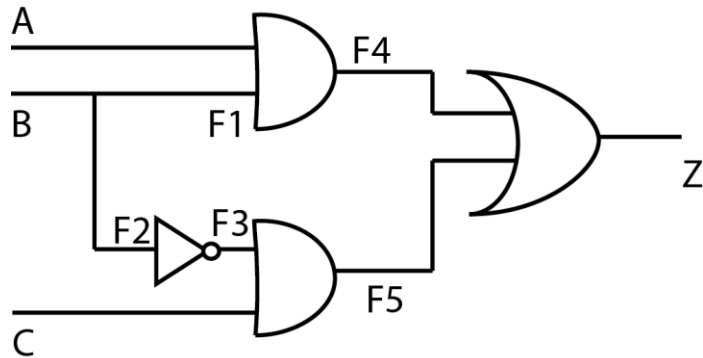


Multi-fanout net ($N > 1$):

- $(N+1)$ fault sites

Possible Stuck-at Faults: Illustration

- Consider the circuit alongside
- How many single stuck-at (0/1) faults are possible in this circuit?

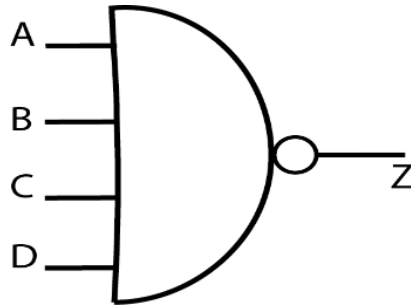


Answer: 18

Detecting a Fault: Test Vectors

Test Vectors

- Any input pattern, or sequence of input patterns, that produces a different output response for a faulty circuit and a fault-free circuit

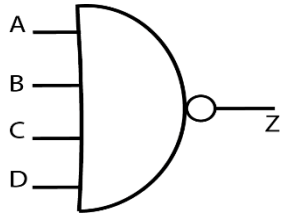


Exhaustive Testing for functionality

- Exponential number of test vectors required (2^N) to test the functionality

Single stuck-at fault model makes the number of faults linear in the number of circuit elements

Design for Testability: Test Vectors



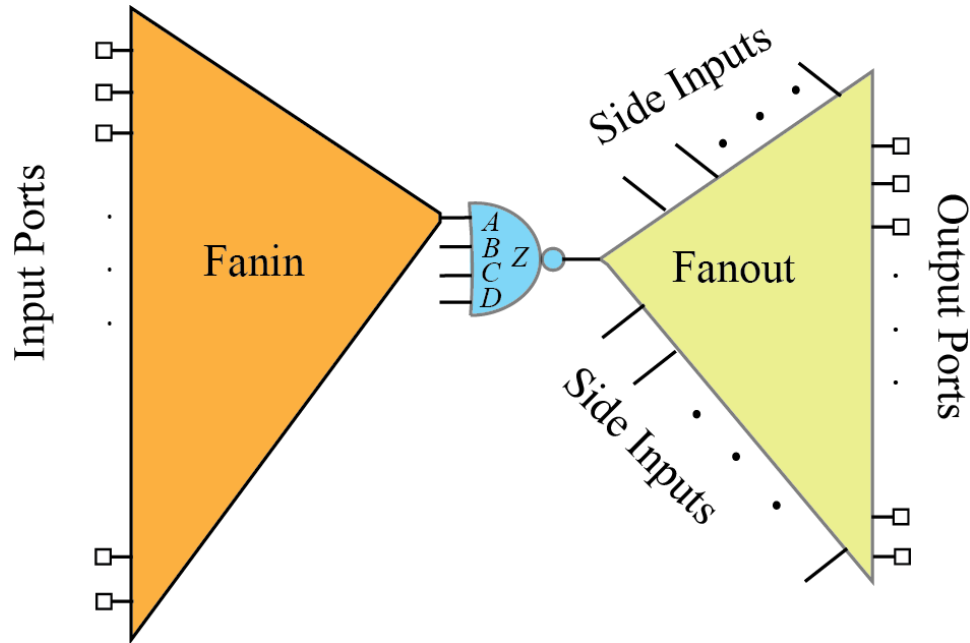
Test Vectors

- 1111 (A/0, B/0, C/0, D/0, Z/1)
- 0111 (A/1, Z/0)
- 1011 (B/1, Z/0)
- 1101 (C/1, Z/0)
- 1110 (D/1, Z/0)

*Number of test
vectors = $n + 1$*

A	B	C	D	Z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Combinational Circuit: Controllability and Observability



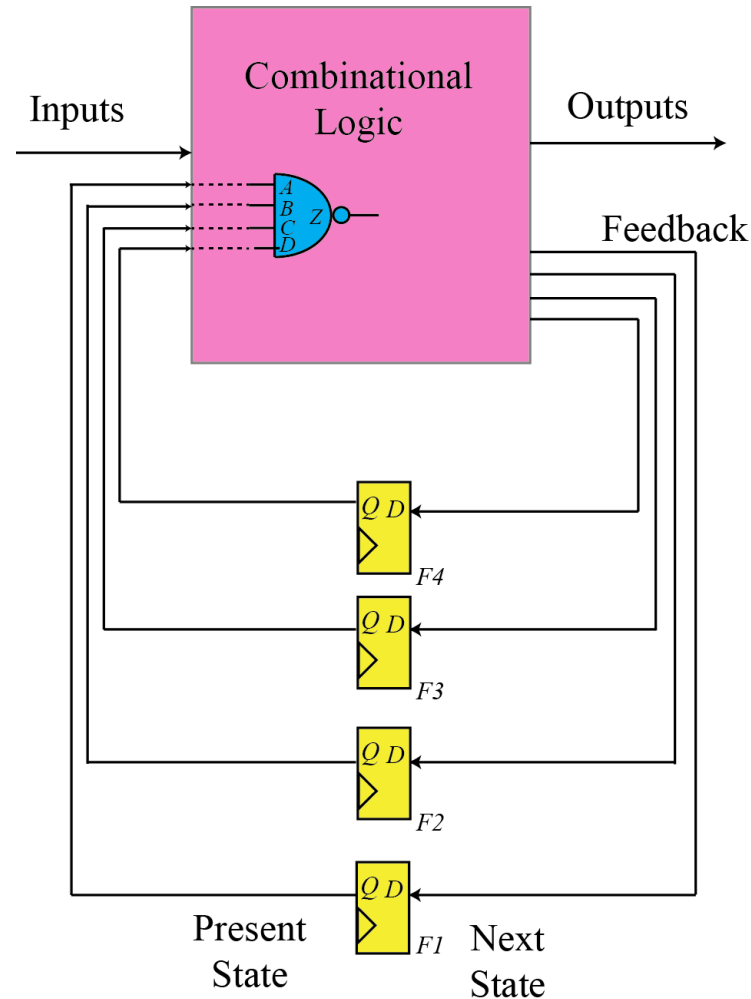
- If the NAND gate was lying too deep in a circuit, it is difficult to apply the required test vectors at the inputs

Controllability: ability to set any desired value (0 or 1) on the internal signals of a circuit by applying an appropriate test-vector to the primary inputs.

- Similarly the output of NAND gate will be difficult to observe at any primary output if the NAND gate is lying too deep in circuit

Observability: ability to examine any internal signal by propagating its value to a primary output by applying appropriate test vector to the primary inputs.

Sequential Circuit: Controllability and Observability

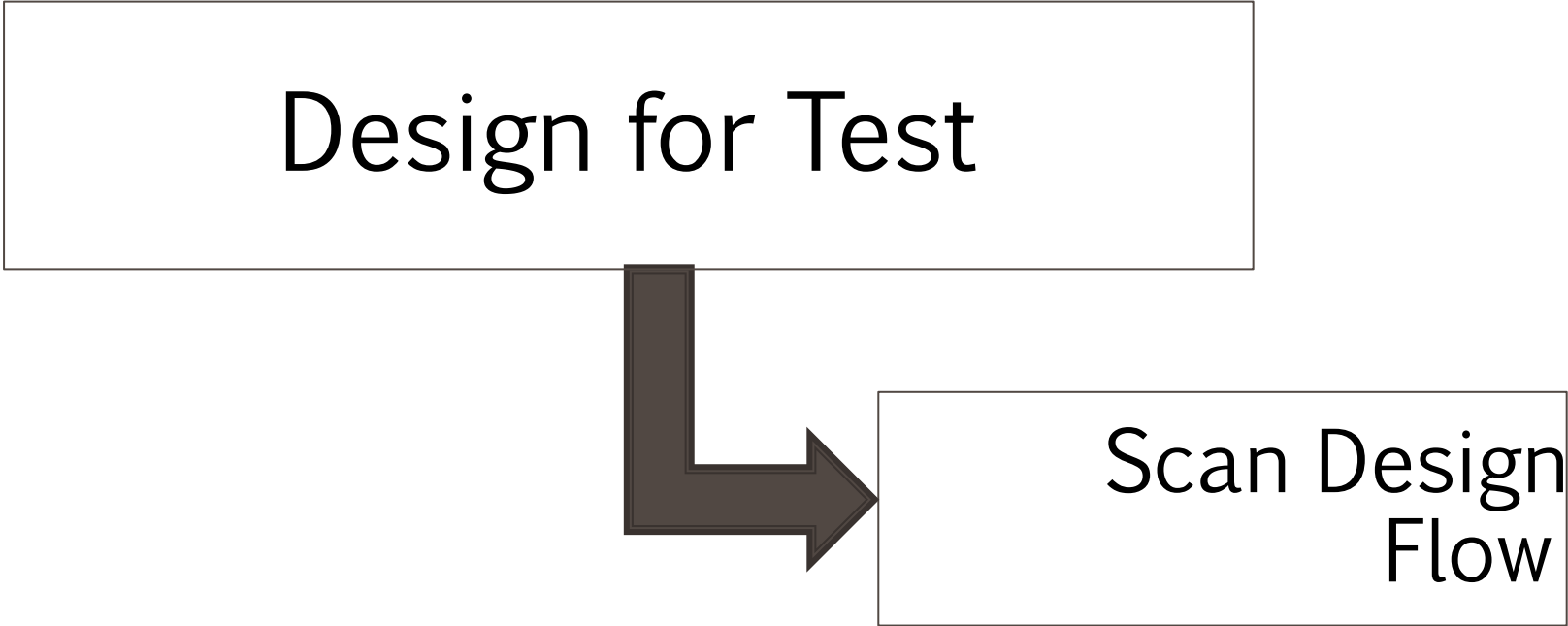


Problem of controllability

- Setting a particular value at any pin in a sequential circuit is more difficult
 - Several cycles may be required to write a particular value
 - State traversal required
 - Finding such a test sequence is time consuming for sequential ATPG tool

Problem of observability

- Similar difficulty in examining the value at a particular pin in the sequential circuit (requires state traversal)

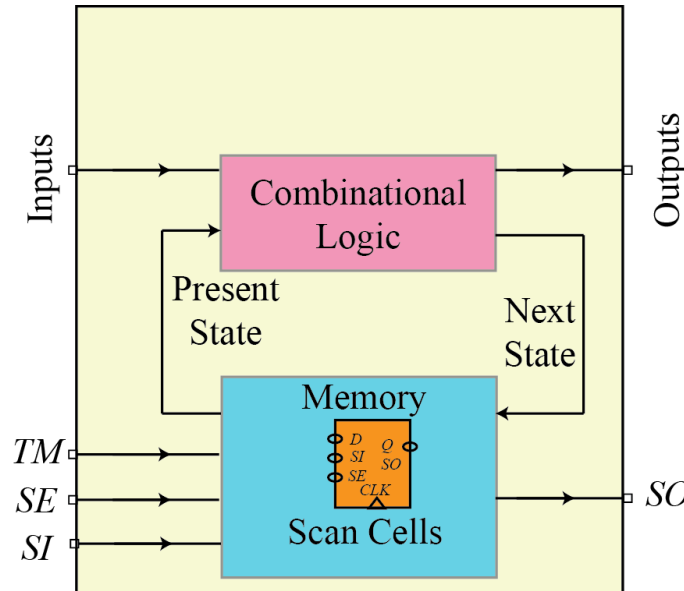
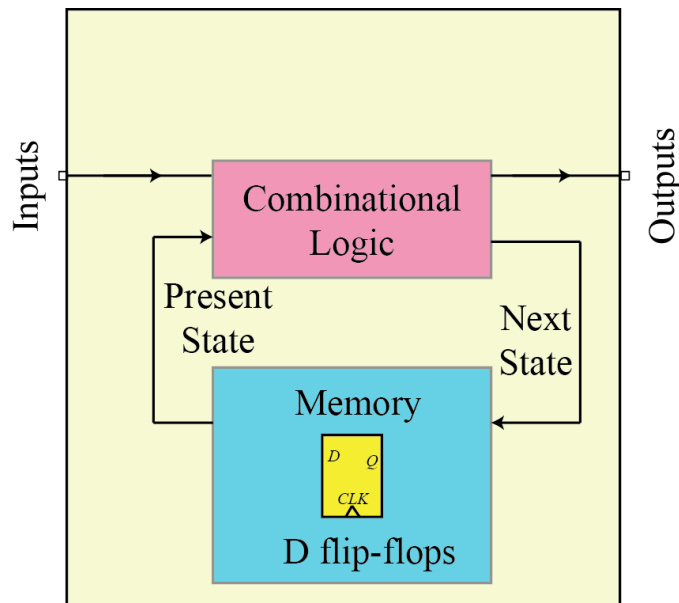


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graph TD; A[Design for Test] --> B[Scan Design Flow]
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Design for Test

Scan Design
Flow

Scan Design Flow: Design Modifications



- Extra primary ports added
 - TM (Test Mode)
 - SE (Scan Enable)
 - SI (Scan In)
 - SO (Scan Out)

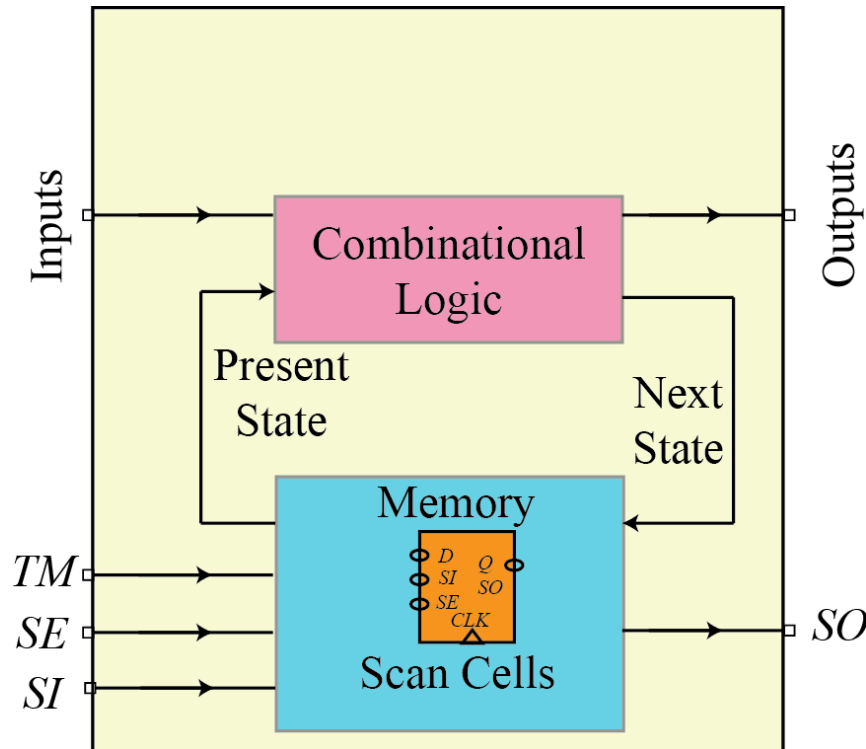
- D flip-flops replaced with another memory elements (**scan cells**)

- Scan cells are reconnected to form shift register (**scan chain**)

Effect

- Dramatically improve *controllability* and *observability* of memory elements (flip-flop) in a sequential circuit

Scan Design Flow: Different Modes



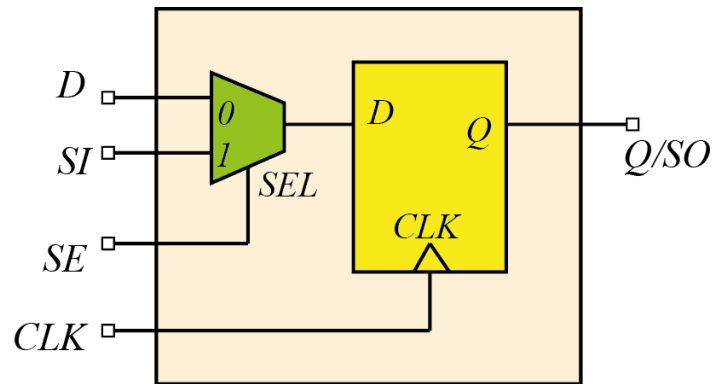
Design works in three modes:

1. **Normal Mode:** functional mode in which chip works
2. **Shift Mode:**
 - Memory elements (i.e. scan cells) work as shift registers
 - Test vectors are shifted-in and responses are shifted-out
3. **Capture Mode:** response of the fabricated circuit is captured during testing

Mode	TM	SE
Normal	0	0
Shift	1	1
Capture	1	0

Scan Cells

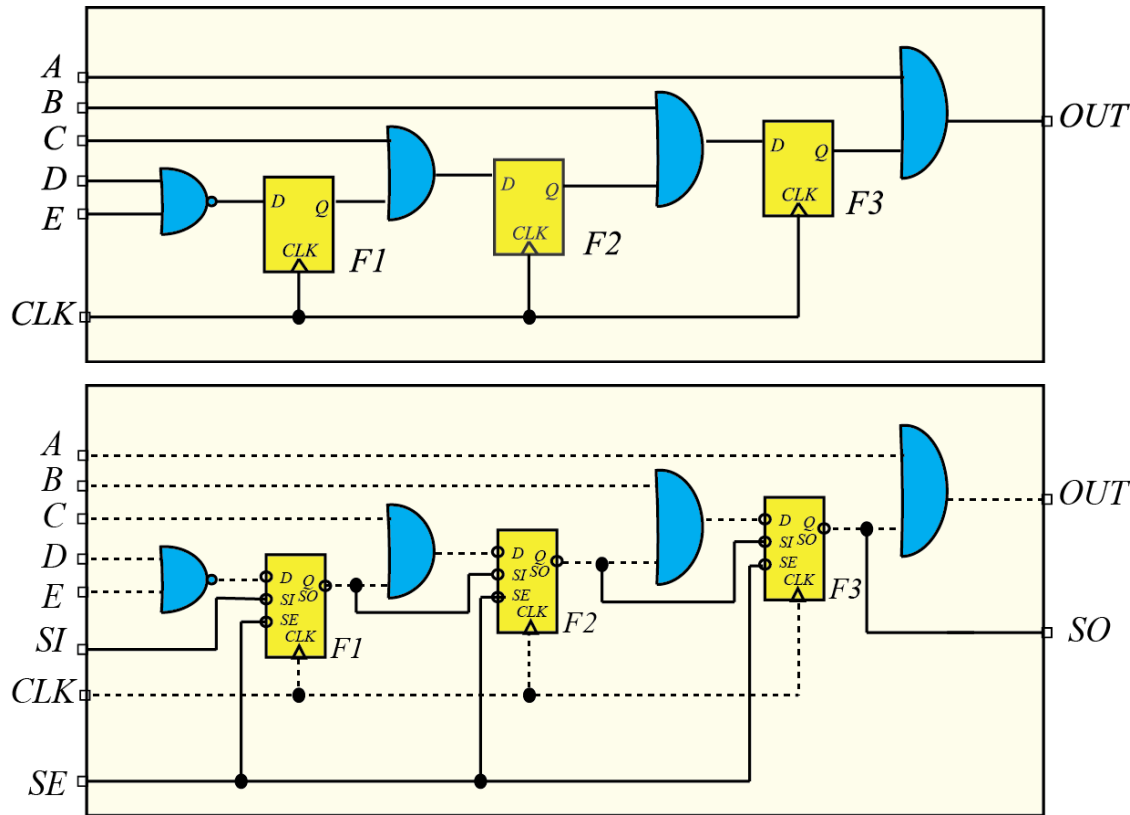
- Different kinds of scan cell can be used
- Most popular is MUXED-D Scan Cell



- D, CLK, Q similar to D flip-flop
- SI $\equiv$ scan input
- SE $\equiv$ scan enable
- Q also works as SO $\equiv$ scan out

- The multiplexer selects data between D and SI using the value at SE pin
- In the normal/capture mode: SE=0
 - Value at D is latched
- In the shift mode, SE=1
 - Value at SI is latched
- Output pin produces the content of D flip-flop
 - Next State could be of D-pin or SI-pin

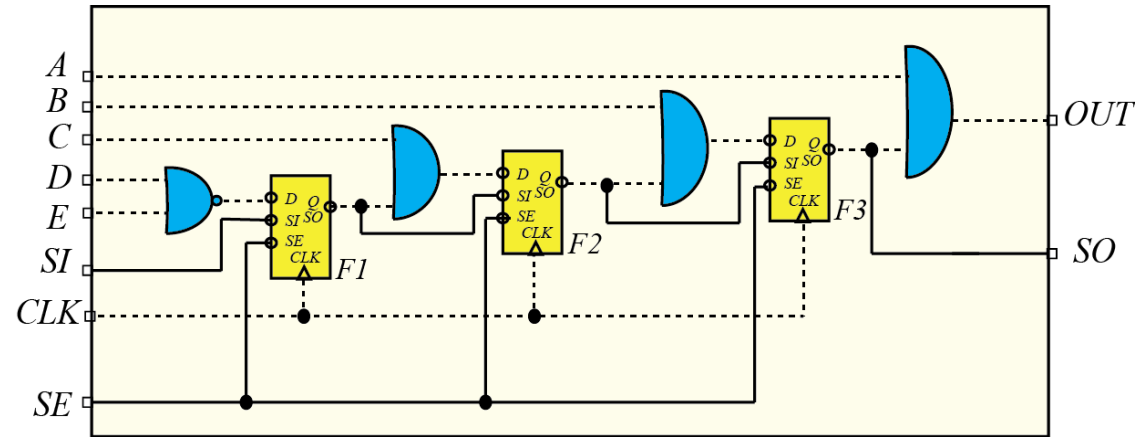
Scan Design Flow: Forming Scan Chain (Example)



- No changes made to the previous connections of D, Q and CLK pins

- The SI pin of the first scan cell is connected to the SI port
- Q/SO pin of one cell is connected to SI pin of the next cell to form a chain
- Q/SO pin of the last scan cell is connected to the SO port
- All SE pins of scan cells are connected to the SE port
- Form a scan chain consisting of N scan cells
- Any test vector can be shifted-in from SI port in N clock cycles
- Any test response can be shifted-out to SO port in N clock cycles
- Without scan chain, it could take exponential number of cycles

Scan Design: Sequential to Combinational Circuit Testing

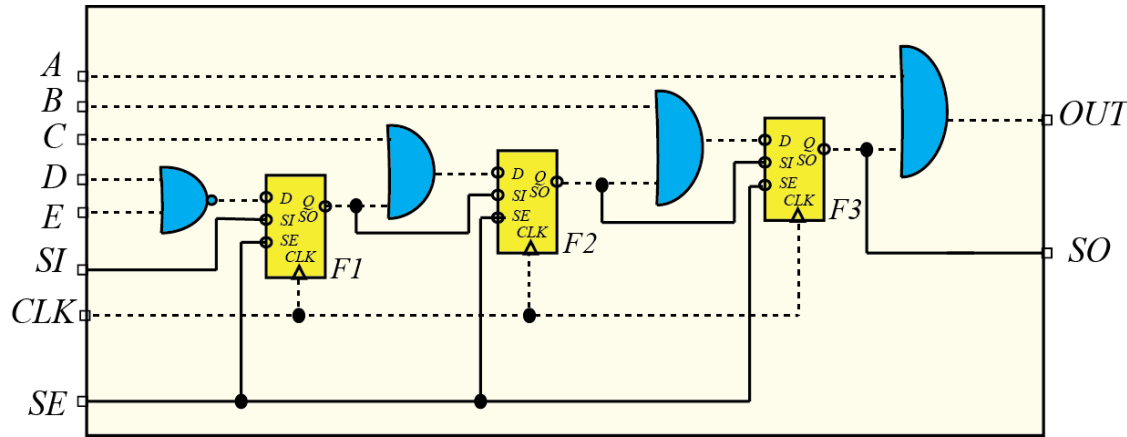


- Pins of a flip-flop become controllable/observable from primary input/output
 - Q-pin can be treated as pseudo-primary input (PPI)
 - D-pin can be treated as pseudo-primary output (PPO)

Scan design eases of testing:

- Effectively transforms the problem of sequential circuit testing to combinational circuit testing
- Automatic test pattern generation (ATPG) problem effectively changes from sequential to combinational

Scan Design Flow: Mechanism



Shift Mode: set SE=1.

- Shift in the desired test vector using port SI to the scan cells F1, F2, F3.
- Apply the required test vector at input port also.

Capture Mode: set SE=0 for 1 clock cycle.

- If there was a fault for which test vector was applied, scan cells will capture the result of fault and receive “fault” output

Shift Mode: switch back to shift mode (SE=1)

- Shift out the captured result to the port SO.
- The result is compared with the expected response
- Simultaneously, apply next test vector at port SI and allow it to scan-in through the scan chain.

Scan Design Flow: Tasks

Design Preparation: Design becomes testable

- Guidelines that must be followed during designing such that scan design flow can be used effectively

Scan Synthesis: Design becomes Scan Design

- **Scan Configuration** (During Synthesis)
 - Decide number of scan chains, scan cells to be used, exclude certain elements from being converted to scan cells
- **Scan Replacement** (During Synthesis)
 - Replace Flip-Flops with Scan Cells
- **Scan Reordering and Stitching** (During Physical Design)
 - Reorder scan cells based on physical location so that routing becomes easier

- **Test Vector Generation**
- **Scan Verification**
 - Scan Shift/Capture Operation using logic simulator
 - Verify Timing (STA): Hold Violations

Scan Design Flow: Cost

- Area Overhead: Scan Cell, Routing resource
- IO Pin Cost
- Performance degradation: added delay of multiplexor
- Design Effort Cost

Scan Design Flow: Number of cycles required

Problem:

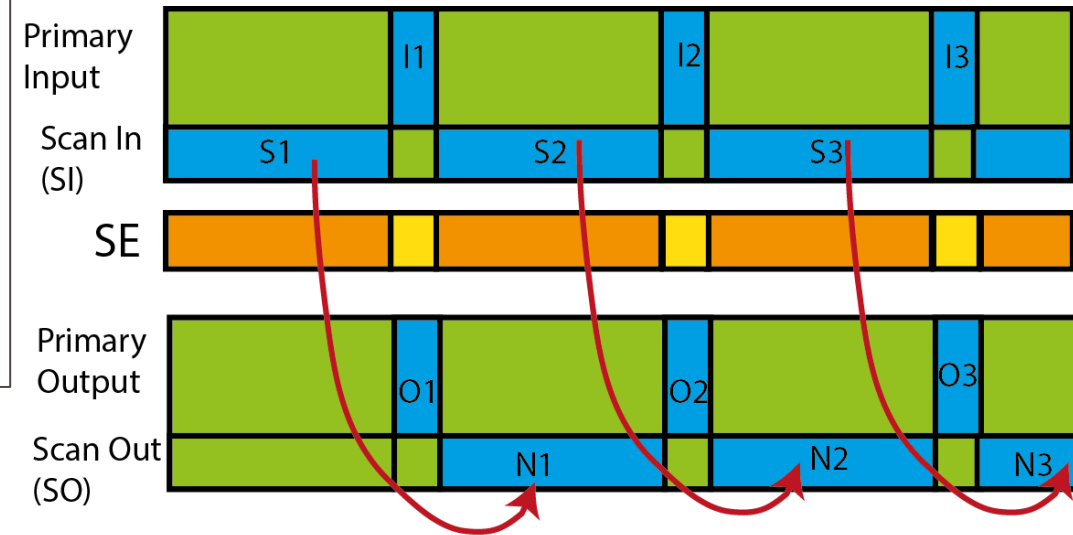
Given:

Number of Test Vectors = T

Number of Scan Cells in Scan Chain = N

Calculate:

Number of Clock Cycles Required.



Test Bits (Input
or Output)

Don't Care or
Random Bits

0

1

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.